Inorganic Cumulative Exam

April 24, 2015

Electron-Transfer Reactions: Electron-transfer reactions are central to many of the most intensely studied current topics in chemistry. Therefore, a general working knowledge of the field is important and including what is popularly called "Marcus Theory" after R. A. Marcus (Nobel Prize 1992).

Marcus theory rests on some very simple concepts

1. Describe the outer-sphere reorganizational energy
2. Describe the inner-sphere reorganizational energy
3. Describe the inverted region
4. Henry Taube (Nobel Prize 1983) published an investigation of the reaction of Cr^{2+} with $Co(NH_3)_5Cl^{2+}$ that very elegantly showed that this reaction was an "inner-sphere reaction". Using the differences in the lability of ligands on metals in various oxidation states describe the mechanism of this reaction focusing on the fate of the Cl^-. You do not need to know what "inner-sphere" means to answer the question but the answer will tell you what it means.

5. A related topic is "intervalence transfer" . Describe this process.
6. Describe an oxygen atom transfer reaction and give an example.
7. Marcus theory rests on the notion that the reactants interact only in a very minor way prior to reaction. What would this tell you if you tried to apply Marcus Theory to the reaction in question six.
8. Marcus theory also rests on the fundamental differences between motion of electrons and the motion of nuclei. How does this fit into the picture?